

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) building algorithms for RTL synthesis and ML-guided digital logic optimization. Built SiliconCrew — an ML-driven RTL synthesis pipeline (Yosys → OpenROAD → GDSII) where LLM agents guide optimization decisions, scoring **37/92 on CVDP** and **5/5 on ASU Spec2Tapeout** — and a from-scratch Static Timing Analysis engine in Python performing graph-based traversal and critical-path extraction over circuit IR.

TECHNICAL SKILLS

Logic Synthesis & EDA: Yosys | OpenROAD | Synopsys Design Compiler | Cadence Innovus | Voltus | SDC | NLDLM Liberty | ASAP7 7nm | SkyWater 130nm

RTL & Verification: SystemVerilog | Verilog | Cocotb | iverilog | SymbiYosys | Static Timing Analysis | Pipelining | Hazard / Forwarding Logic

ML / C++ / Systems: Python | C++ | gem5 (C++ SimObject) | CUDA | LangGraph | Graph Neural Networks | Graph Algorithms | Docker

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

May 2026

Tempe, AZ

Coursework: VLSI Design · VLSI Design Automation · Advanced Computer Architecture · Algorithm/Hardware Co-Design · SoC Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — ML-Guided RTL Synthesis & Optimization Pipeline

Nov 2025 – Present

- Prototyped and evaluated ML methods (LangGraph agent, 21 tools) to guide optimization decisions across an end-to-end synthesis pipeline — RTL generation → lint → simulation → Yosys synthesis → OpenROAD place-and-route → GDSII — across SkyWater 130nm and ASAP7 7nm PDKs, with SymbiYosys formal verification and Cocotb constrained-random; FastAPI + WebSocket backend, MCP server.
- Closes the synthesis-to-signoff loop without human intervention: on simulation failure, reads VCD waveforms, diagnoses signal-level bugs, patches RTL, and re-verifies; enforces SDC constraint validation, Yosys RTL-vs-netlist equivalence checking, and timing-signoff guardrails before GDSII tape-out.
- Achieved **37/92 first-pass on CVDP** (Comprehensive Verilog Design Problems) and **5/5 on ASU Spec2Tapeout** with zero human intervention — demonstrating ML-driven multi-stage logic optimization successfully integrated into a production-grade synthesis flow.

GCN ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm

Spring 2025

- Designed an 11-module GCN ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and completed the full RTL-to-GDSII flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus APR with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved 50,300 μm^2 chip area at 50.6% core density, 52.29 mW total power (64.8% switching), and 99.2 ns end-to-end latency.

5-Stage Pipelined MIPS Processor — SystemVerilog

Fall 2025 – Spring 2026

- Implemented a 5-stage pipelined MIPS processor in SystemVerilog (33 instructions) with a 12-path forwarding network (EX-to-EX, MEM-to-EX, WB-to-ID, branch, JR) and 6 hazard stall conditions; extended with an 8-entry fully-associative CAM instruction cache (3-state FIFO FSM) and LL/SC atomic operations with link-address tracking.

Static Timing Analysis Engine — Graph-Based Circuit Traversal in Python

Fall 2025 – Spring 2026

- Built a complete STA engine in Python from scratch performing graph-based RTL traversal and analysis — Liberty NLDLM parser (7×7 delay/slew LUTs with bilinear interpolation), .bench circuit-graph parser, topological sort via Kahn's algorithm, forward arrival-time/slew propagation, backward RAT/slack computation, and critical path extraction; validated against ISCAS benchmarks c17, b15, and c7552.

LRU-IPV Cache Replacement Policy — Modern C++ inside gem5

Fall 2024

- Implemented LRU-IPV replacement policy as a first-class SimObject inside gem5's modern C++ codebase — navigated templated class hierarchies and cache interface contracts to integrate 4 interface methods (reset, touch, getVictim, invalidate) across 5 source files; validated across 3 L2 configurations (128/256/512 kB), observing L2 replacements collapse from 637 → 112 → 2 at 41.8M instructions, 2.04 IPC.

WORK EXPERIENCE

Systems Engineer

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.